

Tableaux.mth: Semantic Tableaux for Propositional Classical Logic with DERIVE

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Automated theorem proving (ATP) for Propositional Classical Logic is an algorithm that checks the validity of a formula.

The kind of exercises that can be solved using an ATP can be grouped in the following three Logic problems:

1. Checking the satisfiability of a formula (SAT). Satisfiability is the problem of determining if there exists an interpretation that assigns TRUE to the given formula.
2. Checking the validity of a formula (TAUT). A formula of propositional logic is a tautology (or it is valid) if the formula itself is TRUE for all possible interpretations.
3. Checking the correctness of an inference, that is, determining if a formula (conclusion) can be logically deduced from a set of formulae (hypothesis).

In this paper the file Tableaux.mth is presented. This file allows us to use DERIVE as an ATP for Propositional Classical Logic by means of the Semantic Tableaux algorithm.

The use of this utility file becomes specially useful for teachers and students participating in courses related to Computational Logic within Computer Science degree. In particular, the file can be used as a didactical tool supporting the process of teaching and learning of automated logic deduction.

We will show how to solve problems related to validity, satisfiability and deduction using this utility file.

On the other hand, in order to use DERIVE as a PeCAS (Pedagogical CAS), a graphical approach of the execution of the main algorithm can be optionally shown.

Finally, we will describe the conclusions and the future work on this issue.